MASTER CLASS DELIVERABLE • VIDEO #11

The 100% Free Local AI Master Guide: Run 70B & DeepSeek Offline

Ollama, 4-Bit GGUF Quantization, Open WebUI & Zero Cloud API Invoices

RUNTIME: OLLAMA + LLAMA.CPP MODELS: DEEPSEEK-R1 / LLAMA 3.3 / QWEN 2.5 PUBLISHED BY: @PARKER.RENDER

1. Why Run Models Locally?

Cloud LLM APIs (OpenAI, Anthropic) bill you per token, throttle high concurrency, and transmit sensitive client IP over third-party servers. With **Ollama + GGUF**, you get unmetered, zero-cost inference running completely offline on your workstation GPU or unified memory.

2. Hardware Sizing & Recommended Quants

VRAM / Unified RAM	Recommended Model	Quantization	Tokens/sec
8 GB VRAM	DeepSeek-R1-Distill-Qwen-8B	Q4_K_M	35-45 t/s
16 GB VRAM	Qwen 2.5 Coder 14B / DeepSeek 14B	Q4_K_M	25-35 t/s
24 GB - 32 GB	DeepSeek-R1 32B / LLaMA 3.3 70B	Q4_K_S / IQ3_M	15-22 t/s
64 GB+ (Mac M-series)	LLaMA 3.3 70B / DeepSeek-R1 70B	Q4_K_M	12-18 t/s

3. 3-Minute Setup Script (Windows / Mac / Linux)

```
# 1. Install Ollama
curl -fsSL https://ollama.com/install.sh | sh
# On Windows: Download installer from https://ollama.com/download

# 2. Pull & Run DeepSeek-R1 Distill or LLaMA 3.3
ollama run deepseek-r1:8b
# For coding tasks:
ollama run qwen2.5-coder:14b

# 3. Launch Dockerized Open WebUI (ChatGPT clone locally)
docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway \
  -v open-webui:/app/backend/data --name open-webui --restart always \
  ghcr.io/open-webui/open-webui:main
```

4. Python Integration via OpenAI-Compatible API

```
from openai import OpenAI

client = OpenAI(
    base_url="http://localhost:11434/v1",
    api_key="ollama" # Required by client, ignored by Ollama
)
```

```
response = client.chat.completions.create(
    model="deepseek-r1:8b",
    messages=[{"role": "user", "content": "Write an async web crawler in Python."}]
)
print(response.choices[0].message.content)
```

Crafted by **@parker.render** • Visit parker-render.netlify.app/local for updates.